Control Centre

Six panels that set up, run and score one test of a trading model on past prices, read left to right. Each saves its own settings, the Scoreboard runs the test with all of them, and every result is kept with the settings that made it.

Does any rule make money after trading costs on a year of prices the model never saw?

ANSWER SO FAR

Crypto, no. Of 120 random crypto test setups run on 24 September, 9 made money after costs once overlapping trades were removed, and hourly, 4-hour and daily bars all lost at the median.

US stocks, mostly the market. Daily stock setups made money in 43 of 45, but buying every stock made money in all 45, a median 0.71 per cent a trade, and the model's picks added a median of only 0.18.

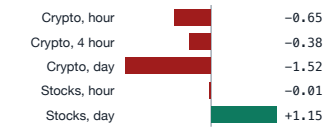
One strategy passed. 12-1 momentum on US stocks cleared every test and has run on a paper account since 18 August.

A1 Data

which prices, and what counts as a win

Chooses the market, the bar size and the outcome the model is trained to predict.

On the board	Default
Market Binance coins, or US stocks from Alpaca	`crypto`
Timeframe 137 coins from LINK on, no bitcoin; for trying settings, not judging a coin	`4 hours, test sample`
Outcome a win when the take-profit comes before the stop; three-way scores up, down or flat instead	`barrier`
Take-profit, Stop fixed when the dataset was built; these fields only check it	`+2, −1 ATR`
Horizon two days on 4-hour bars	`12 bars`



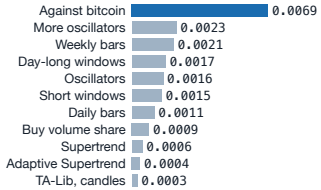
Median after-cost return a trade on the model's top fifth, overlapping trades removed, 200 random setups, 24 September. Costs were 0.20% for crypto and 0.10% for stocks.
`random-suite-20260924-092016.md`

Too few wins
On 4-hour bars the take-profit came first on 25 to 26 per cent of bars, short of the 33.3 per cent that +2 against −1 needs to break even.
`univariate-20260921-021638.md`

A2 Indicators

the columns the model may use

Chooses which families of computed columns the model is given. None ticked offers them all.



Mean rise in the Brier score when a family's columns were shuffled. 4-hour bars, 40,000 training rows, 8 September.
`feature-report-20260908.md`

Bitcoin leads
Strength against bitcoin carried three times the weight of the next family, and it is the only family that reads another asset's price.

Useless columns
16 of 90 columns changed the error by less than 0.00001 when shuffled, 9 of them candle patterns.

INDICATOR ENGINES

MACD, moving averages, Fibonacci levels and the confluence score only draw this panel's charts. The model never reads their settings, and none has been tested on prices it had not seen.

B1 Variables

which columns are kept

Can drop weak columns before the fit, with a penalised regression on the training years only. It is off by default.

On the board	Default
Screen first on runs the selection before the fit	`off`
L1 mix 1 is the lasso, which drops columns, 0 is ridge, which only shrinks them, and between is the elastic net	`1.0`
Penalty rule the strongest penalty within one standard error of the best	`1se`
Rows training rows the selection is fitted on	`25,000`

61 columns offered
46 kept at the best penalty
0 kept at one standard error

Lasso on 25,000 rows of the 1-hour dataset, 21 September.
`variable-selection-20260921-0218.md`

Not yet measured
Nothing survived the default rule, and no record compares the test-year score with and without the selection.

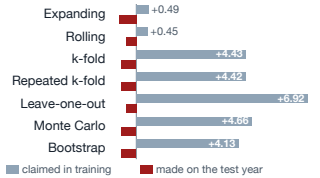
Single columns
Fitted alone on 4-hour bars, 16 of 33 columns cleared p < 0.05, against about 2 expected by chance.
`univariate-20260921-021638.md`

B2 Training

how the past is split

Holds back the last year for one final test, and cuts the years before it into folds.

On the board	Default
Blind days the test year, scored once at the end	`365`
Regime the resampling method, walk-forward on a growing window	`expanding`
Folds cuts of the training years	`3`
Purge bars 0 to 48 tested; error moved by 0.0001	`0`



After-cost return a trade, in per cent, each method claimed for the model it chose, against that model on the test year. Crypto, 4-hour bars.
`blind-length-study-20260924-053206.md`

Shuffled splits overstate
Methods that ignore time order claimed 4.1 to 6.9 per cent a trade, and the models they chose lost 0.41 to 0.59 per cent on the test year.

Short tests mislead
Split into twelve 30-day stretches, the same test year named 3 to 7 different models as best.

C1 Scoreboard

fit, score and compare

Chooses the models, runs the test, and ranks the result against every fit on record. A fit passes with Theil's U2 below 1 on the test year and an RMSE ratio at or under 1.1; a three-way run is scored on money after cost.

On the board	Default
Models fits three models, logistic regression, random forest and HistGBM, not all six	`none ticked`
Class weight lifted test U2 to 1.14 to 1.16 on every model that takes it; no weighting did better	`balanced`

Six models on 3 coins, no class weight, 16 September. Red is above the 1.1 limit.

Model	Test U2	RMSE ratio
Logistic, elastic net	0.995	1.010
Logistic regression	0.998	1.021
Random forest	1.002	1.073
Classic boosting	1.006	**1.180**
HistGBM	1.092	**4.379**
LightGBM	1.093	**5.487**

Tuning overfitted
Tuning HistGBM cut training RMSE from 0.354 to 0.074 while held-out RMSE rose from 0.484 to 0.517.
`model-tuning-histgbm-20260908.md`

The record
574 fits, 119 with test U2 below 1, the best 0.980. Four repeats of one setup spread by 0.0062.

C2 Ledger

the paper account, in dollars

Draws the figures a run asks for and reports the paper stock account.

12-1 momentum passed
Hold the 50 US stocks with the best return over the past year, leaving out the latest month. Over 116 months the top tenth beat the market by 1.01 per cent a month (t = 2.41).
`monthly-factors-20260905.md`



The paper account against the same $100,000 in SPY, the S&P 500 fund, at daily closes.

Marked 15 October, 16:15 New York

Account, from $100,000	$96,382
Change since 18 Aug	−3.6%
SPY, same period	+0.6%
45 open positions	+$1,222
78 closed trades	−$4,852
Worst holding, AXTI	−15.2%
Cost a fill, 5 to 10 bp assumed	8.4 bp

Paper money
No real money moves, and the live switch is off. The app's scans have left 7,255 paper tickets since 25 September, 3,468 settled, and 9 user runs have left 22.

Testing an idea

1 Change one thing
Change one setting on its panel and save it. Two changes at once cannot be told apart.

2 Run the test
Press Run the test on C1, which uses every saved setting.

3 Read the marks
Test U2 below 1, RMSE ratio at or under 1.1, and for the three-way outcome, money after cost.

4 Compare
Against the run before. A change in U2 smaller than 0.0062 is within the spread of repeats.

5 Keep or revert
Keep what beat the record. Never choose a setting by looking at the test year, because it has then been seen.

Terms

ATR
Average true range, the typical high-to-low move over 14 bars.

RMSE
Root mean squared error. Held-out over training RMSE is the RMSE ratio, or overfit ratio.

Triple barrier
A trade ends at the take-profit, the stop or the time limit, whichever comes first.

Top fifth
The fifth of bars the model rated most likely to rise.

Base rate
The share of bars where the take-profit came first.

Brier score
The mean squared error of a stated probability.

Test year
The final 365 days, held back and scored once, the board's blind period.

MACD
A fast moving average of price minus a slow one.

Fold
One cut of the training years, fitted on part and scored on the rest.

Gradient boosting
Trees fitted in turn, each to the last one's errors. HistGBM and LightGBM are two kinds.

Walk-forward
Fit on the past, score the stretch after it, then move forward in time.

Basis point
One hundredth of a per cent, written bp.

Theil's U2
The model's error over that of always forecasting the base rate. Below 1 beats it.

Paper account
Real prices and simulated orders, with no money at risk.

Using the App

The public page at seamusrobertmurphy.github.io/swing-trader. Anyone sets up a model, runs it on fresh prices and gets paper trades settled on real prices, with no keys, no code and no money.

How do I run a model, change it, and read the trades it makes?

WHAT CHANGED

26 Sep Presets on every page, a Binance or Alpaca switch, and the cost floor.
2 Oct The Forecast card, with paper trades, TimesFM fans and Why this call.
14 Oct Candles, Supertrend bands, volume, RSI, exit tools and the Indicator boards.
15 Oct A daily refresh, three trial strategies, the weekly scoreboard and stock settling.

1 Start

Quick start, home page and A1

A model runs from a phone in about five minutes, and nothing is bought or sold.

1. Tap Binance for crypto or Alpaca for US stocks.
2. Tap a preset, which fills every setting on every panel.
3. Type your name and press Run Model, on the home page, B2 or C1.
4. Open C2 for your paper trades. Each is settled on real prices when its time is up.

Preset	Holds a trade
Best on record random forest on 4-hour candles, the lowest validation error that passed the overfit check	`2 days`
Three-way outcome calls each daily candle up, down or flat, the most validation profit after cost	`24 days`
Quick and simple elastic-net logistic regression on 4-hour bitcoin, ether and solana, trained in time order	`2 days`
Any preset on stocks daily candles of three stocks, +3 against −1 ATR	`20 trading days`

Stocks chosen on the test year
Each stock preset trades the three of 20 stocks its model did best on in its test year, so only the trades settling from now on are a fair test.

The Glossary link beside Quick start defines every term on the page.

2 Settings

configuration tools, A1 to C1

Each setting has a one-line note under it. A1 is open, and A2, B1, B2 and C1 fold theirs under Advanced, model building. Saved settings are what Run Model sends.

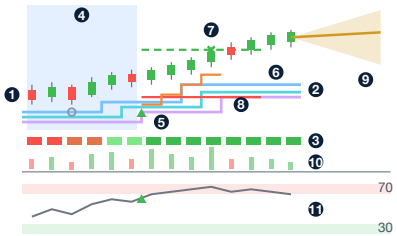
Block	Panel
Choose Market market, candle length, and the coins or stocks	A1
Choose Basket a quick pick, the symbols and how much history to load; Explore draws their prices	A1
Choose Filter volatility band, cost floor, relative volume and money flow at entry	A1
Choose Ranking what to rank by, and keep only the top third	A1
Choose Label take-profit, stop, horizon, win or loss against up, down or flat, and the break-even band	A1
Choose Features the indicator families the model may read	A2
Choose Screen drop weak columns with a lasso before the fit	B1
Regime and split cross-validation design, folds, repeats, test days and embargo	B2
Model, settings, grid learners, class weight, overfit cap, each learner's settings and a grid search	C1

On A1 a setting's pictures appear under it once you change it.

3 Forecast chart

trades on candles, home page and C2

Shows each preset's own paper trades on recent candles. Pick Binance or Alpaca, a preset, then a symbol tab.



A drawing of the marks, not real prices.

1. Candles, green when the close beat the open, red when not.
2. Three Supertrend bands, trailing stops under the price in an uptrend; a line breaks where its band flips.
3. Agreement strip, how many bands point up, red for none to green for all three.
4. Blue shading, the 24 candles before a taken trade.
5. Green triangle, a trade taken; a grey ring is one skipped.
6. Take-profit in green dashes, stop in red.
7. A cross where it closed, green at take-profit, red at stop, grey at the time limit.
8. Trailing stop, in orange, when its button is on.
9. Gold fan, TimesFM's forecast now, 10th to 90th percentile; blue fans are its forecasts at taken trades.
10. Volume, green on an up candle.
11. Lower panel, MACD or RSI with its 70 and 30 zones, and each taken trade's reading.

4 Exits

lower panel and exit tools

LOWER PANEL

MACD by default, a fast average of the price minus a slow one. RSI switches to the 14-candle relative strength index, 0 to 100, above 70 after a fast rise and below 30 after a fast fall, with each taken trade's RSI marked.

TRAILING STOP

Replays the research's exit on every taken trade, a stop two volatility widths under each close that only steps up and a take-profit that shrinks to the entry price by the time limit. A diamond marks where it would have closed.

EXIT SHAPES

Under the table, one taken trade closed three ways side by side, fixed, trailing, and trailing with a shrinking take-profit. Touch a green triangle to choose it.

THE TABLE

Under the chart, it sets taken trades against skipped ones on result, exit reason, bands up at entry, run-up, volume and RSI, and the trailing exit against the fixed one.

Touch any mark
Every candle, trade and line gives its numbers. How to read it opens the full note, and Key the legend.

5 Indicator boards

three engines on daily candles

The workflow's three engines, run unchanged on daily candles of the Forecast card's symbols, on the market the Binance or Alpaca switch chose. Pick a board, then a symbol from its top-left menu.

Board	Marks
MACD triangles are crossings large enough to trust, diamonds a divergence of price and averages	▲ ◆
Confluence marks a trade when enough of MACD, moving averages, candle pattern and Fibonacci position agree; its title gives that rule against holding	`buy, sell`
Fibonacci where a pullback in the latest large swing often stops; 50 to 61.8 per cent is the golden pocket, dashed lines are targets	`levels`

Charts only
The model does not read these boards, and their rules have not been tested on prices they had not seen.

WHEN IT UPDATES

1. Scans every four hours, at 23 minutes past.
2. Due trades settled hourly, at 17 minutes past.
3. Forecasts and boards rebuilt daily from 00:30 UTC.
4. Old page history wiped on the 1st of each month, current files kept.

6 Results

C2, your trades and the scoreboard

Paper trades lists every user's runs, newest first, yours highlighted, under three charts.

Taken against skipped plots the running total after cost, so the gap between its lines is the edge. Model scores gives each run's RMSE, MAE, Theil's U2 or overfit ratio, and What carries the book the total by model or symbol.

Each trade's Why this call gives its score group, expected move and which of the four checks passed.

Weekly scoreboard on crypto, in the Performance Log. Average per cent after cost of a taken trade, and the count.

Week closed	Best	Quick
28 Sep	`-1.09 (3)`	`none`
5 Oct	`-0.40 (4)`	`none`
12 Oct	`+0.74 (2)`	`-1.60 (1)`

Crypto three-way and all stock preset trades are still open; the first stock ones close on 26 October.

TRIAL STRATEGIES

Best on record with one change each, begun 15 October. Shorter hold halves the holding time, looser buy level drops the crypto break-even check, and trend filter needs all three bands up.

No edge yet
User runs' 3 settled taken trades averaged −1.31 per cent after cost, the 9 skipped +1.28. No strategy has the 30 trades the confirmed-best rule needs.

Revising a design

1 Start from a preset
Tap the nearest preset so every other setting is known.

2 Change one thing
One setting on A1 or under Advanced. Two changes at once cannot be told apart.

3 Run Model
About five minutes. Your name marks your trades on C2.

4 Wait for settling
Crypto 4-hour trades close in 2 days, stock trades in 20 trading days. An open trade has no result.

5 Compare
Taken trades should beat skipped ones, and the weekly scoreboard sets the presets and trials on the same weeks.

Terms

Paper trade
A trade recorded at a real price with no money, also called a ticket.

Settled
Closed at its take-profit, stop or time limit, with a result after cost.

Taken, skipped
Whether the model's call passed all four checks.

Supertrend band
A trailing stop that flips side when the trend turns.

Run-up
The price move over the 24 candles before an entry.

RSI
Relative strength index, recent gains against losses, 0 to 100.

TimesFM
Google's forecasting model, drawn as a fan and never traded on.

Trailing stop
A stop that follows the price up and never down.

Shrinking take-profit
A target that falls to the entry price by the time limit.

Cost floor
Drops the costliest fifth of candles to trade.

Edge floor
The least expected move after cost a taken trade needs.

Trial strategy
Best on record with one change, a shorter hold, a looser buy level or a trend filter.

Weekly scoreboard
Each strategy's average taken trade by the week it closed.

Confirmed-best rule
30 independent trades with an interval above zero, fixed before any settled.